2 System wiring overview

Every assembly on one page, joined by real wires:
battery to motors, the safety chain, rails, ESP32,
I2C chain. Drawing only; nets are on pages 3–9.

File: overview.kicad_sch

3 Car harness and box edge

Everything outside the box: battery + F1, key, pedal,
shifter, 5 motors, E–stop, sensors, dash, speakers.
Box plugs J1–J12 (J1–J8 mate the stock plugs, TBD #17).

File: harness.kicad_sch

4 Power and safety chain

K1 contactor and its coil chain: key, E–stop (2 NC),
Q1 hold MOSFET. F2 key feed, fuse block F3–F8,
ground star TB1, motor bus TB2, battery divider.

File: power.kicad_sch

5 Low–voltage rails

U1 LM2596 5V_LOGIC, U2 AMS1117 3V3_SENSE,
U3 5V_ACC and U4 12V_ACC for accessories.

File: rails.kicad_sch

6 Drive, variant A (4 + 1 IBT–2)

U10–U13 IBT–2 per wheel, U14 steering IBT–2.
470 µF each, enable pull–down per board,
IS current–sense networks to the ADS1115s.

File: drive.kicad_sch

7 Controller and inputs

ESP32 (nets from include/pins.h), ELRS receiver,
pedal input protection, wheel–speed pull–ups,
DS18B20, DFPlayer serial.

File: controller.kicad_sch

8 I2C bus and switch inputs

2 x ADS1115 current sense, PCF8574 switches
(shifter, horn, E–stop sense, seat), MPU–6050,
PCA9685 accessory PWM.

File: i2c_io.kicad_sch

9 Accessories: fans, lights, sound

MOSFET board for lights and fans, 12 V branch
fuses F9/F10, music amp, DFPlayer + alert speaker.

File: accessories.kicad_sch

## How to read it

Every pin has a stub and a net label: the label names the net, so each pin says where it goes. Labels with a flag shape are global (used on more than one sheet); plain text labels stay on one sheet. Each box above is a sheet: double–click it in KiCad to open it (or page down in the PDF). Location field: box, A1 (the interface board), car, dash. Source field: owned, ordered 24 Oct, stock car, to buy, TBD. Tables: NETS.md, CONNECTORS.md, WIRE_LIST.md, BOM.md (same data).

## Three safety layers (POWER_CHAIN.md)

1. A power chain a person can always cut, no software:
   F1, key SW1 and E–stop S2 (two NC blocks) in the K1 coil.
2. Drivers that default off: DRIVE_EN / STEER_EN pulled down
   at every board; K1 coil MOSFET Q1 pulled down.
3. The ESP32 only adds control on top, and can only open K1.

## Open items (TBD)

#17 stock plug types and pin use: J1–J8 (car–facing plugs)
#33 motor pairing in the harness: J6/J7 splitter or not
#13 battery voltage and chemistry: D2 TVS value, BTS7960 27.5 V limit
#14 motor labels: per–motor current, variant A or B (#32)
#18 box size: variant A (5 boards) or B (2 boards)
#16 pedal type: switch, pot or 5 V hall (divider)
#15 IBT–2 IS resistor: config IS_RESISTOR_OHMS
Key switch rating (SW1 carries ~8 A worst case)
Shifter: signal–only or motor current (only signal can go to U24)
K1 precharge resistor: after the bench test
AS5600 cable length: I2C or analog OUT on U23 A1
Variant B current sensors (#32)

## Revisions

0.3 2026–10–24  First KiCad version, from draft v0.2 (SVG).
Net names, connector/pin tables, wire list, BOM
matched to the 24 Oct order. Changes are listed
in README.md, 'Changes from draft v0.2'.

Draft v0.3, not for building the car–facing side until the TBDs close

**RideOnESP32 · Can–Am Maverick X3 24 V 4WD**

Sheet: /
File: rideonesp32.kicad_sch

***Title: RideOnESP32 v3 wiring schematic (index)***

| Size: A3 | Date: 2026–10–24 | **Rev: 0.3** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 1/9 |

# System wiring overview: everything connected

Each thin line is one wire; thick blue, green and grey lines are labelled bundles. Wire numbers, gauges and every single conductor are in WIRE_LIST.md; pins in CONNECTORS.md. Drawing only: the nets live on sheets 3–9.

CAR SIDE (stock parts and our off-box parts)

THE BOX (in the stock controller's place under the seat)

MOTORS (stock)

BT1 battery
24 V, chemistry TBD (#13)
+
-

W001 10 AWG

F1 40 A
mr15 cm from + b
a
b

J1 BATT

10 AWG

K1 contactor
125 A, 24 V coil
A
C+
C−

K1 → TB2, 10 AWG

TB2 MOTOR_24V

F2 15 A
inline
a
b

J2 KEY

SW1 key
stock dash
1
2

key_24v
KEY

FB1 fuse block
bus = KEY_24V
IN
F3
F4
F5
F6
F7
F8

coil → K1

S2 E-STOP
latching, 2 NC in series + 1 NO
NC in
NC out
NO
NO gnd

J9 E_STOP

J3 PEDAL

Pedal
type TBD (#16)
V+
SIG
GND

J4 SHIFT

Shifter F/R
stock, signal only
F
R
GND

J12 → TB1 GND star, 12 AWG

J8 ACC

Horn button, lights
stock (TBD #17)
L+
L−
HORN
GND

J10 SENS

Sensors
H1/H2 A3144 at the wheels, U25 AS5600 on the steering column, SW5 seat
5V
GND
WS_L
WS_R
3V3
SCL
SDA
SEAT

5V_KY-DATA

J11 DASH

Dash
U27 OLED, LS1 alert speaker
3V3
GND
SCL
SDA
SPK+
SPK−

echo: 5V to OLED + speaker, 3V3 to sensors

J12 MUSIC

Music speakers
LS2, LS3 (+ tweeters)
L+
L−
R+
R−

U2 AMS1117 3.3 V
3V3_SENSE
out
in

U1 LM2596 5 V
5V_LOGIC
in
out
gnd

U3 buck 5 V 5 A
5V_ACC
in
out
gnd

U4 buck 12 V
12V_ACC
in
out
gnd

3V3

U20 ESP32
pins from pins.h
VIN/GND
FL
FR
RL
RR
ST
analog
GPIO12
SCL/SDA
RX/TX

PWM_12KHZ_EN_325 (GPIO)

IS_FL
IS_FR
IS_RL
IS_RR

CONTACTOR

MOTOR_sense_analog

U10 IBT-2 FL
BTS7960, 470 µF on B+/B−
ctrl
IS
B+
B−
M+
M−

U11 IBT-2 FR
BTS7960, 470 µF on B+/B−
ctrl
IS
B+
B−
M+
M−

U12 IBT-2 RL
BTS7960, 470 µF on B+/B−
ctrl
IS
B+
B−
M+
M−

U13 IBT-2 RR
BTS7960, 470 µF on B+/B−
ctrl
IS
B+
B−
M+
M−

J6 FRONT

J7 REAR

M1 FL
100 nF on the tabs
+
−

M2 FR
100 nF on the tabs
+
−

M3 RL
100 nF on the tabs
+
−

M4 RR
100 nF on the tabs
+
−

J6/J7: one pair per motor. If the stock harness pairs motors, a reversible splitter at the motor plugs (#33).

A1 interface board
Q1, IS RC + clamps, pedal and hall inputs, divider, buses
Q1 drain
pedal
halls
buses out
IS out
GND
5V in
3V3 in
to ESP32
5V/3V3 out
IS in

A1 ground bus → TB1

F9 / F10
3 A lights, 10 A amp
in
ilights
amp

12 V lights

lights low side

U21 ELRS receiver
CRSF, antenna outside
5V
UART

U14 IBT-2 STEER
BTS7960, 470 µF on B+/B−
ctrl
IS
B+
B−
M+
M−

J5 STEER

M5 steering
100 nF on the tabs
+
−

5V_ACC

5V_KY-DATA

SCL/SDA

I2C to J10 / J11

IS filtered

FAN1/2 24 V
4box fans
−

U29 MOSFET board
low-side switches
in
fan
ilights

U30 DFPlayer
Car sounds
pwr
spk

PWM

I2C bus

U22
ADS1115 0x48
IS x5
in
out

U23
ADS1115 0x49
in
out

U24
PCF8574 0x20
in
out
P0–P7

U26
MPU-6050
in
out

U28
PCA9685
in
out
PWM

U32 music amp
ZK-502T, Bluetooth here only
in
out

alert speaker

music L/R

TB1 GND star

Legend

- Battery + before K1, 10 AWG
- MOTOR_24V after K1, 10 AWG
- Power ground to TB1, 10 AWG
- Motor leads, 12 AWG
- Key-switched 24 V branches (F2–F8)
- K1 coil chain (F3 1 A)
- 5 V
- 3.3 V
- 12 V accessories
- Driver control / PWM bundles
- I2C bus
- Analog: IS, pedal
- Switches, sensors, serial
- Logic ground

Draft v0.3, not for building the car-facing side until the TBDs close

**RideOnESP32 · Can-Am Maverick X3 24 V 4WD**

Sheet: /2 System wiring overview/

File: overview.kicad_sch

**Title: System wiring overview**

| Size: A2 | Date: 2026-10-24 | Rev: 0.3 |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 2/9 |

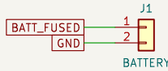
*Left: what stays in the car (stock parts and our off-box parts). Right of each plug: into the box.*
*Plugs facing the stock harness (J1–J8) are TBD until the stock plug photos (#17).*
*J9–J12 are our own sealed plugs.*

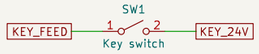
## Battery and key

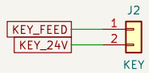




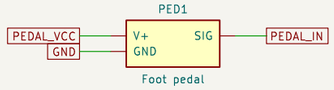


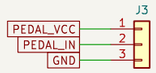



## Pedal and shifter

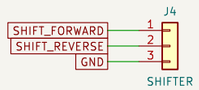


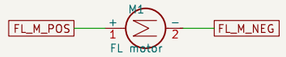







## Front motors



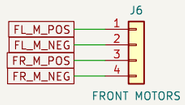




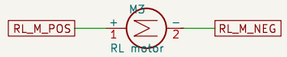








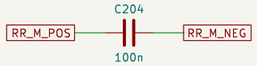

## Rear motors

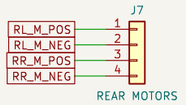








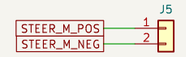



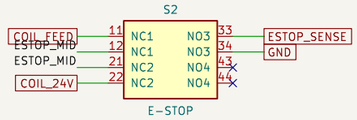

## Steering motor, E-stop

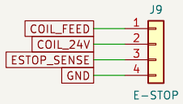




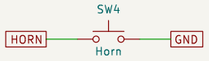




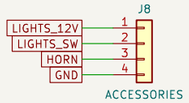




## Horn and lights







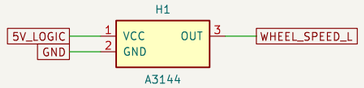


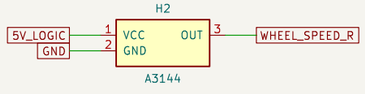

### Going back to stock

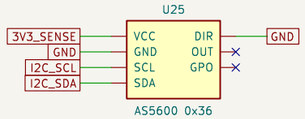
1. Key off. Unplug J1–J8 from the box.
2. Plug the stock HY2005RX controller into the same plugs (motor splitters: unplug, rejoin).
3. Unplug J9–J12; the stock car has none of them.

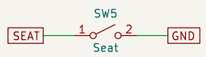
No stock wire is ever cut (#33).

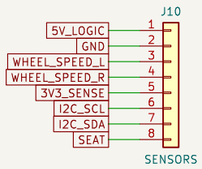
## Sensors



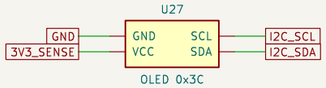



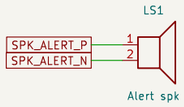



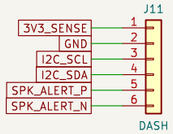



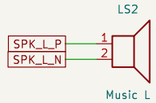

## Dash

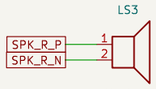


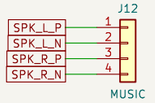




## Music




Draft v0.3, not for building the car-facing side until the TBDs close

**RideOnESP32 · Can-Am Maverick X3 24 V 4WD**

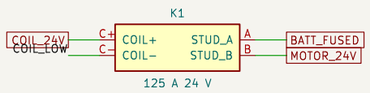
Sheet: /3 Car harness and box edge/
File: harness.kicad_sch

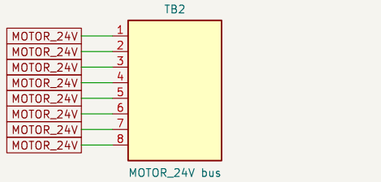
***Title: Car harness and box edge***

| Size: A3 | Date: 2026-10-24 | **Rev: 0.3** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 3/9 |

*Layer 1, no software: key and E-stop in series with the K1 coil; either one opens K1 whatever the ESP32 does.*
*Q1 lets the ESP32 open K1 but never close it. Reset, crash or brown-out opens it (R2 pull-down).*
*Rails come from the key side, before K1, so the ESP32 stays up after an E-stop and can report why.*

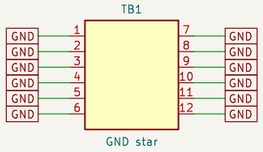
## Main path



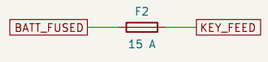



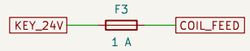






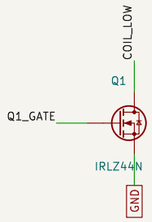

## Coil chain











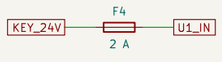


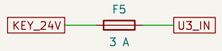



### Coil chain

KEY_24V -> F3 1 A -> J9.1 -> S2 NC1 -> S2 NC2
-> J9.2 -> K1 coil -> Q1 -> GND.
Unplugging J9 opens K1 (fail-safe).
Q1 gate: GPIO12 via R1, R2 pull-down.

## Fuse block (key side)



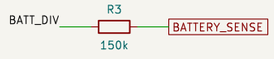



















## Notes

### Main current path

BT1 + -> F1 40 A -> J1 -> K1 stud A -> K1 stud B
-> TB2 -> B+ of U10-U14 (10 AWG).
B- of U10-U14 -> TB1 star -> J1 -> BT1 -.
The blade fuse block carries only the small
branches; it is not in the 40 A path.
K1 precharge: decide after the bench test
(5 x 470 µF behind a 125 A contactor).

### Fuse rule

Every fuse protects the wire after it:
the wire is rated above the fuse.
F1 40 A: 10 AWG. F2 15 A: 14 AWG.
F6 10 A: 16 AWG. F3-F5, F7, F8: 20 AWG
or heavier. See WIRE_LIST.md.
















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**RideOnESP32 · Can-Am Maverick X3 24 V 4WD**

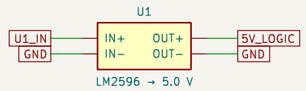
Sheet: /4 Power and safety chain/
File: power.kicad_sch

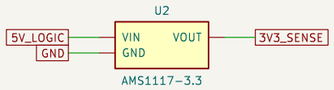
***Title: Power and safety chain***

| Size: A3 | Date: 2026-10-24 | **Rev: 0.3** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 4/9 |

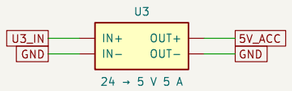
*Logic and accessories never share a converter. 3V3_SENSE comes from 5V_LOGIC, not the ESP32's own 3.3 V pin.*
*Set each buck's output with a meter before connecting any load.*

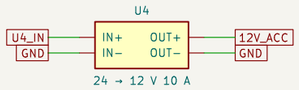
## Logic




## Accessories




## Budget

### Rail budget

| | | | | |
|---|---|---|---|---|
| 5V_LOGIC | U1 LM2596 | ESP32, RX, halls, U2 | ~0.5 A | F4 2 A |
| 3V3_SENSE | U2 AMS1117 | IBT-2 logic, I2C | ~0.1 A | (via F4) |
| 5V_ACC | U3 5 A | DFPlayer, USB, LEDs | ≤5 A | F5 3 A |
| 12V_ACC | U4 10-20 A | lights F9, amp F10 | ≤10 A | F6 10 A |

Fuses are on the 24 V input side: 5 A at 5 V is about 1.2 A at 24 V.

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**RideOnESP32 · Can-Am Maverick X3 24 V 4WD**

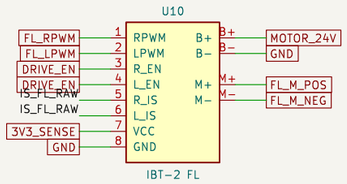
Sheet: /5 Low-voltage rails/
File: rails.kicad_sch

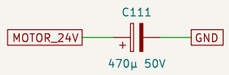
***Title: Low-voltage rails***

| | | |
|---|---|---|
| Size: A3 | Date: 2026-10-24 | **Rev: 0.3** |
| KiCad E.D.A. 9.0.9 | | Id: 5/9 |

*Layer 2: DRIVE_EN and STEER_EN are pulled down at every board, so boot, reset, flashing and crashes leave the drivers off.*
*IS sense: R_IS + L_IS joined (only one side conducts at a time) -> 4.7k -> ADS1115, 10 nF filter, clamp to 3.3 V.*
*Variant B (2 x MDD20A) replaces this sheet: see variant_b/.*

## FL (front-left)

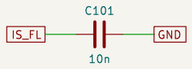






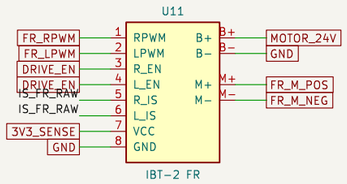


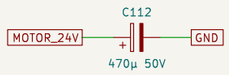







## FR (front-right)

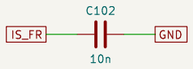






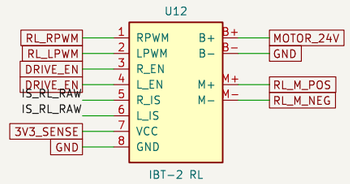


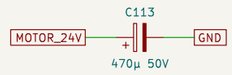







## RL (rear-left)

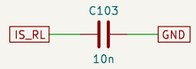






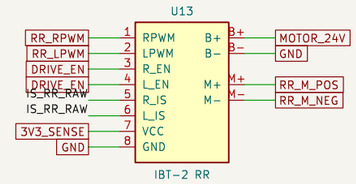


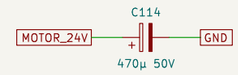







## RR (rear-right)

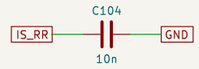






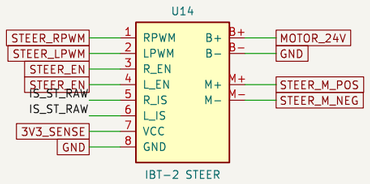


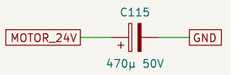







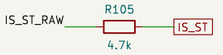
## STEER (steering)

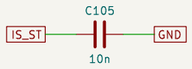












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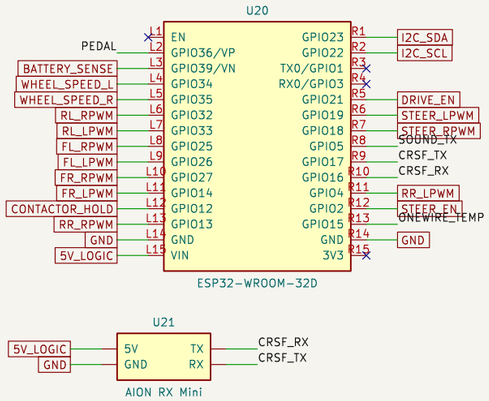
Sheet: /6 Drive, variant A (4 + 1 IBT-2)/
File: drive.kicad_sch

***Title: Drive, variant A (4 + 1 IBT-2)***

| Size: A3 | Date: 2026-10-24 | **Rev: 0.3** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 6/9 |

*Pin nets come from include/pins.h; CI keeps pins.h, PINOUT.md and WIRING.md in step and check_netlist.py checks this sheet.*
*Analog inputs on ADC1 only. GPIO36/39 carry analog only (they glitch when the ADC runs). Kill lines on pins quiet at boot.*

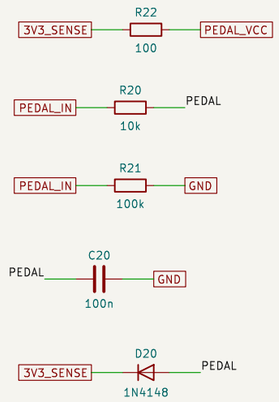
## ESP32







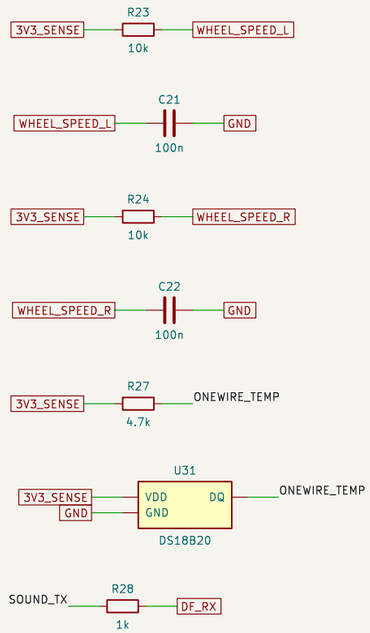
## Pedal




### Pedal input

Switch pedal: between V+ and SIG.
Pot: ends on V+ / GND, wiper on SIG.
5 V hall throttle: add a divider (TBD #16).
Shorted high reads as full pedal: open bench check (CLAUDE.md).

## Wheel speed, temperature, sound




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**RideOnESP32 · Can-Am Maverick X3 24 V 4WD**

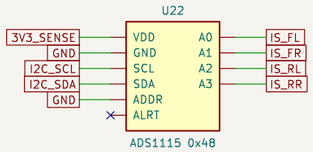
Sheet: /7 Controller and inputs/
File: controller.kicad_sch

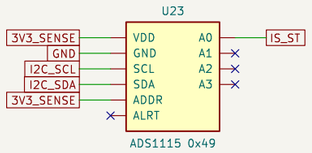
***Title: Controller and inputs***

| Size: A3 | Date: 2026-10-24 | **Rev: 0.3** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 7/9 |

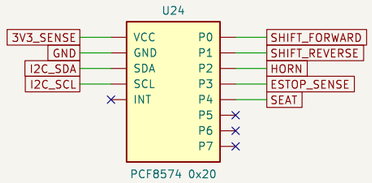
*I2C on GPIO22 (SCL) / GPIO23 (SDA) at 3.3 V. Addresses: 0x20 0x36 0x3C 0x40 (+0x70 all-call) 0x48 0x49 0x68.*
*Off-box I2C (AS5600 on the steering column, OLED on the dash) is the weak spot: keep runs short, twisted with GND (#36).*

## Current sense ADCs



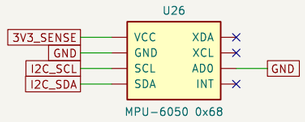

## Switches

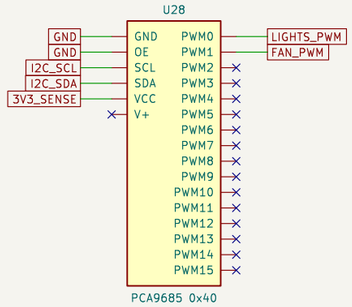



### PCF8574 inputs

P0 shift forward, P1 shift reverse, P2 horn,
P3 E-stop sense (NO contact: 0 = pressed),
P4 seat switch. P5–P7 spare.
The E-stop itself cuts K1 in hardware;
P3 only tells the firmware why.

## IMU, PWM, pull-ups










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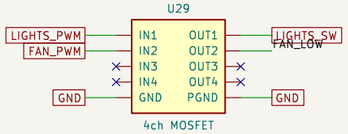
Sheet: /8 I2C bus and switch inputs/
File: i2c_io.kicad_sch

***Title: I2C bus and switch inputs***

| Size: A3 | Date: 2026-10-24 | **Rev: 0.3** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 8/9 |

*Phase 7. Nothing here is in the 24 Oct order except the inline fuse holders.*
*Music has its own amp with Bluetooth on that board only; the ESP32 never runs Wi-Fi or Bluetooth.*

## Switched loads

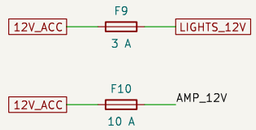


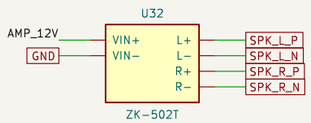







## 12 V branches

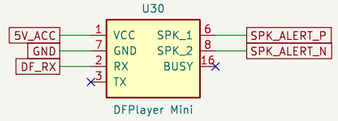






## Car sounds




### Audio

DFPlayer on 5V_ACC so speaker current stays
off the ESP32 supply. See docs/hardware/AUDIO.md.

Draft v0.3, not for building the car-facing side until the TBDs close

**RideOnESP32 · Can-Am Maverick X3 24 V 4WD**

Sheet: /9 Accessories: fans, lights, sound/
File: accessories.kicad_sch

***Title: Accessories: fans, lights, sound***

| Size: A3 | Date: 2026-10-24 | **Rev: 0.3** |
|---|---|---|
| KiCad E.D.A. 9.0.9 | | Id: 9/9 |